Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/19/2022
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	Effect of Space Radiation on Innate Immune System Homeostasis Drives Immune and Endothelial Cell Dysfunction and Neurodegeneration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/01/2021	**End Date:**	01/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	No changes have been made		
COI Name (Institution):	Calvi, Laura M.D. (University of Rochester) Love, Tanzy Ph.D. (University of Rochester) Olschowka, John Ph.D. (University of Rochester) Williams, Jacqueline Ph.D. (University of Rochester)		

Grant/Contract No.:	80NSSC21K0542
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The key objective of this proposal is to evaluate dose- and time-dependent effects of simulated space radiation on multiple tissue endpoints associated with normal aging. These endpoints include measures of the immune system, in particular those cells in the blood and tissues that represent a first line of defense for reacting to injury, the vascular compartment, another critical component of tissue homeostasis, and the brain, a critical system that shows age-associated degeneration which also occurs after radiation exposure. These studies will be carried out with male and female mice exposed to simulated space radiation at the NASA Space Radiation Laboratory using modern molecular techniques to measure tissue and cellular changes in gene and protein expression, alterations in cell phenotypes by flow cytometry, and behavioral tests and computer-assisted imaging in brain to quantify degenerative changes. This proposed work responds to Topic 6, Space Radiation and Tissue Homeostasis, in NASA's solicitation, and will provide key information about space radiation effects on the promotion of age-associated changes in multiple systems that are likely tied together in the etiology of late degenerative changes that may impact future astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research focuses on mechanisms of normal tissue injury in brain and bone marrow following radiation exposure. Identification of specific pathways involved in late effect of radiation on these tissues may provide clues for protective or mitigating strategies in the setting of clinical radiotherapy or accidental radiation exposure.
Task Progress:	In this second year of the grant we carried out irradiations during NASA Space Radiation Laboratory-22A (NSRL-22A) for our second and third proposed experiments, which were galactic cosmic radiation (GCR) dose and dose rate studies, respectively. For the GCR dose experiment, we irradiated 72 male and 60 female, 6-month old C57BL/6 mice with 50 cGy 600 MeV/μ iron and 100, 50 or 20 cGy GCRsim BNL (15-18 mice of each sex in each group). An additional 33 mice were similarly processed at the NSRL (e.g., placed in holders for similar times), but not exposed to radiation. For the dose rate experiment, 50 male and 45 female mice were split between 3 groups: 100 cGy GCRsim delivered in 24 fractions, 100 cGy GCRsim delivered acutely, or sham irradiated mice. All mice were handled identically for this expeiment. Following irradiation all mice were shipped back to Rochester for further experiments. We carried out tissue collections from groups of these mice at 11 days post-irradiation (n=5), and behavioral testing, following by tissue collection at 6 months post-irradiation (n= 10 to 15 at each time point). This last time point was completed in late November 2022. Tissue and behavioral data analyses are in progress. We presented an abstract describing the overall goals of our project at the NASA Human Research Program (HRP) Investigators' Workshop (IWS) in February 2022 and published a review article describing the potential roles of microglia in central nervous system (CNS) radiation responses to space radiation.
Bibliography Type:	Description: (Last Updated: 01/07/2026)
Abstracts for Journals and Proceedings	O'Banion MK, Calvi LM, Olschowka JA, Love TMT, Williams JP. "Effects of space radiation on innate immune system homeostasis drives immune and endothelial cell dysfunction and neurodegeneration." 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Articles in Peer-reviewed Journals	O'Banion, MK. "Microglia: Rheostats of space radiation effects in the CNS [central nervous system] microenvironment." Life Sci Space Res. 2022 Aug 9. 2022 Nov;35:180-86. https://doi.org/10.1016/j.lssr.2022.08.002 , Aug-2022
Articles in Peer-reviewed Journals	Zablotska LB, Zupunski L, Leuraud K, Lopes J, Hinkle J, Pugeda T, Delgado T, Olschowka J, Williams J, O'Banion MK, Boice JD, Jr., Cohen SS, Mumma MT, Dauer LT, Britten RA, Stephenson S. "Radiation and CNS effects: Summary of evidence from a recent symposium of the Radiation Research Society." Int J Radiat Biol. 2022 Nov 11:1-11. https://doi.org/10.1080/09553002.2023.2142984 ; PMID: 36318723 , Nov-2022
Articles in Peer-reviewed Journals	Lynch OF, Calvi LM. "Immune dysfunction, cytokine disruption, and stromal changes in Myelodysplastic Syndrome: A review." Cells. 2022 Feb 8;11(3):580. https://doi.org/10.3390/cells11030580 ; PMID: 35159389; PMCID: PMC8834462 , Feb-2022
Articles in Peer-reviewed Journals	Boerma M, Davis CM, Jackson IL, Schaue D, Williams JP. "All for one, though not one for all: Team players in normal tissue radiobiology." Int J Radiat Biol. 2022;1-61. PMID: 34129427; https://doi.org/10.1080/09553002.2021.1941383 ; PMCID: PMC8781287 , Feb-2024